CEUE203 - OOP25AIML034

PRACTICAL – 1

Aim:

1. VendingMachine.java To develop a Java program using an enum and switch expression to accept coins, calculate the total amount, and return the change after purchasing a snack.

2. TollBooth.java To develop a Java program using records, switch expressions, and loops to calculate the total toll collected from vehicles and identify the most frequent vehicle type.

3. RPSLS.java To develop a Java program using enums, switch expressions, random numbers, and loops to implement the Rock-Paper-Scissors-Lizard-Spock game for five rounds and determine the overall winner.

Code Screenshot:

VendingMachine.java

Run

```java

import java.util.Scanner;

public class VendingMachine {

    enum Coin {
        ONE, TWO, FIVE, TEN
    }

    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        int price = 15;
        int total = 0;

        while (total < price) {
            System.out.print("Enter coin (ONE, TWO, FIVE, TEN): ");
            String input = sc.next().toUpperCase();

            Coin coin;

            try {
                coin = Coin.valueOf(input);
            } catch (IllegalArgumentException e) {
                System.out.println("Invalid coin. Try again.");
                continue;
            }

            int value = switch (coin) {
                case ONE -> 1;
                case TWO -> 2;
                case FIVE -> 5;
                case TEN -> 10;
            };

            total += value;
            System.out.println("Total so far: " + total);
        }

        System.out.println("Paid. Change: " + (total - price));

        sc.close();
    }
}

```

```
import java.util.Scanner;


public class TollBooth {


    record Vehicle(String number, String type) {}


    public static void main(String[] args) {
```

```
Scanner sc = new Scanner(System.in);

int total = 0;
int bike = 0;
int car = 0;
int truck = 0;

while (true) {
   System.out.print("Enter vehicle number (or done): ");
   String number = sc.next();

   if (number.equalsIgnoreCase("done")) {
      break;
   }

   System.out.print("Enter vehicle type (bike, car, truck): ");
   String type = sc.next().toLowerCase();

   Vehicle v = new Vehicle(number, type);

   int toll = switch (v.type()) {
      case "bike" -> 20;
```

```
        case "car" -> 50;
        case "truck" -> 150;
        default -> -1;
    };

    if (toll == -1) {
        System.out.println("Invalid vehicle type.");
        continue;
    }

    total += toll;

    switch (v.type()) {
        case "bike" -> bike++;
        case "car" -> car++;
        case "truck" -> truck++;
    }
}

System.out.println("Total toll: " + total);

String mostFrequent = "bike";
```

```
        int max = bike;

        if (car > max) {
            max = car;
            mostFrequent = "car";
        }

        if (truck > max) {
            mostFrequent = "truck";
        }

        System.out.println("Most frequent: " + mostFrequent);

        sc.close();
    }
}
```

3.

```
import java.util.Random;
import java.util.Scanner;

public class RPSLS {
```

```
enum Move {
    ROCK, PAPER, SCISSORS, LIZARD, SPOCK
}

static int winner(Move a, Move b) {
    if (a == b) {
        return 0;
    }

    boolean wins = switch (a) {
        case ROCK ->
            b == Move.SCISSORS || b == Move.LIZARD;

        case PAPER ->
            b == Move.ROCK || b == Move.SPOCK;

        case SCISSORS ->
            b == Move.PAPER || b == Move.LIZARD;

        case LIZARD ->
            b == Move.SPOCK || b == Move.PAPER;
```

```
            case SPOCK ->
                b == Move.SCISSORS || b == Move.ROCK;
        };

        return wins ? 1 : -1;
    }

    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        Random random = new Random();

        Move[] moves = Move.values();

        int playerScore = 0;
        int computerScore = 0;

        for (int round = 1; round <= 5; round++) {
            System.out.println("\nRound " + round);
            System.out.print(
                "Enter ROCK, PAPER, SCISSORS, LIZARD, or SPOCK: "
            );
```

```
Move player;

try {
    player = Move.valueOf(sc.next().toUpperCase());
} catch (IllegalArgumentException e) {
    System.out.println("Invalid move. Try again.");
    round--;
    continue;
}

Move computer = moves[random.nextInt(moves.length)];

System.out.println("Your move: " + player);
System.out.println("Computer move: " + computer);

int result = winner(player, computer);

if (result == 1) {
    System.out.println("You win this round!");
    playerScore++;
} else if (result == -1) {
```

```
            System.out.println("Computer wins this round!");
            computerScore++;
        } else {
            System.out.println("This round is a tie!");
        }

        System.out.println(
            "Score: You " + playerScore +
            " - " + computerScore + " Computer"
        );
    }

    System.out.println("\nFinal Score: You " + playerScore +
                       " - " + computerScore + " Computer");

    if (playerScore > computerScore) {
        System.out.println("You win!");
    } else if (computerScore > playerScore) {
        System.out.println("Computer wins!");
    } else {
        System.out.println("Overall tie!");
    }
```

```
        sc.close();
    }
}
```

Output Screenshot:

Output

```
Enter coin (ONE, TWO, FIVE, TEN): ONE
Total so far: 1
Enter coin (ONE, TWO, FIVE, TEN): FIVE
Total so far: 6
Enter coin (ONE, TWO, FIVE, TEN): TEN
Total so far: 16
Paid. Change: 1
```

Output

```
Enter vehicle number (or done): 1
Enter vehicle type (bike, car, truck): car
Enter vehicle number (or done): 2
Enter vehicle type (bike, car, truck): truck
Enter vehicle number (or done):
```

Output

```
Round 1
Enter ROCK, PAPER, SCISSORS, LIZARD, or SPOCK: SPOCK
Your move: SPOCK
Computer move: LIZARD
Computer wins this round!
Score: You 0 - 1 Computer

Round 2
Enter ROCK, PAPER, SCISSORS, LIZARD, or SPOCK:
```